



ACADEMIC PROJECT PRESENTATION

TOPIC: SOCIAL MEDIA BASED SUICIDE IDEATION

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PROJECT OVERVIEW

In today's digital era, expressing feelings and emotions through social media has become a norm. With the hike in mental health disorders and suicidal attempts globally, this project leverages social media text analysis using NLP, LLMs and crowd annotation, which continuously monitors user activity to detect signs of suicidal risk, enabling timely alerts for intervention and prevention of harm.

Previous Phases of the Project (Phase 1 & Phase 2):

In the previous phases of development, we prepared our customized Crowd Annotated Dataset (100 of 500 samples), resulting in two different outcomes of Annotations from two Groups of Annotators (Phase 1), on which various LLM and NLP models were trained to establish the baseline of our proof of concept (Phase 2).

- Best Model for Suicidal1 (Phase 2): ALBERT (95% Accuracy)
- Best Model for Suicidal2 (Phase 2): BERT (95% Accuracy)

Current Phase of the Project (Phase 3):

Performed Data Augmentation by Crowd Annotation on the remaining 400 samples to produce a final 500 sample annotated dataset, inter-tagger agreement evaluation for pattern diversification, training, evaluating and testing model performance further to establish the proof-of-concept for a potential Hybrid Intelligence (HI) framework.

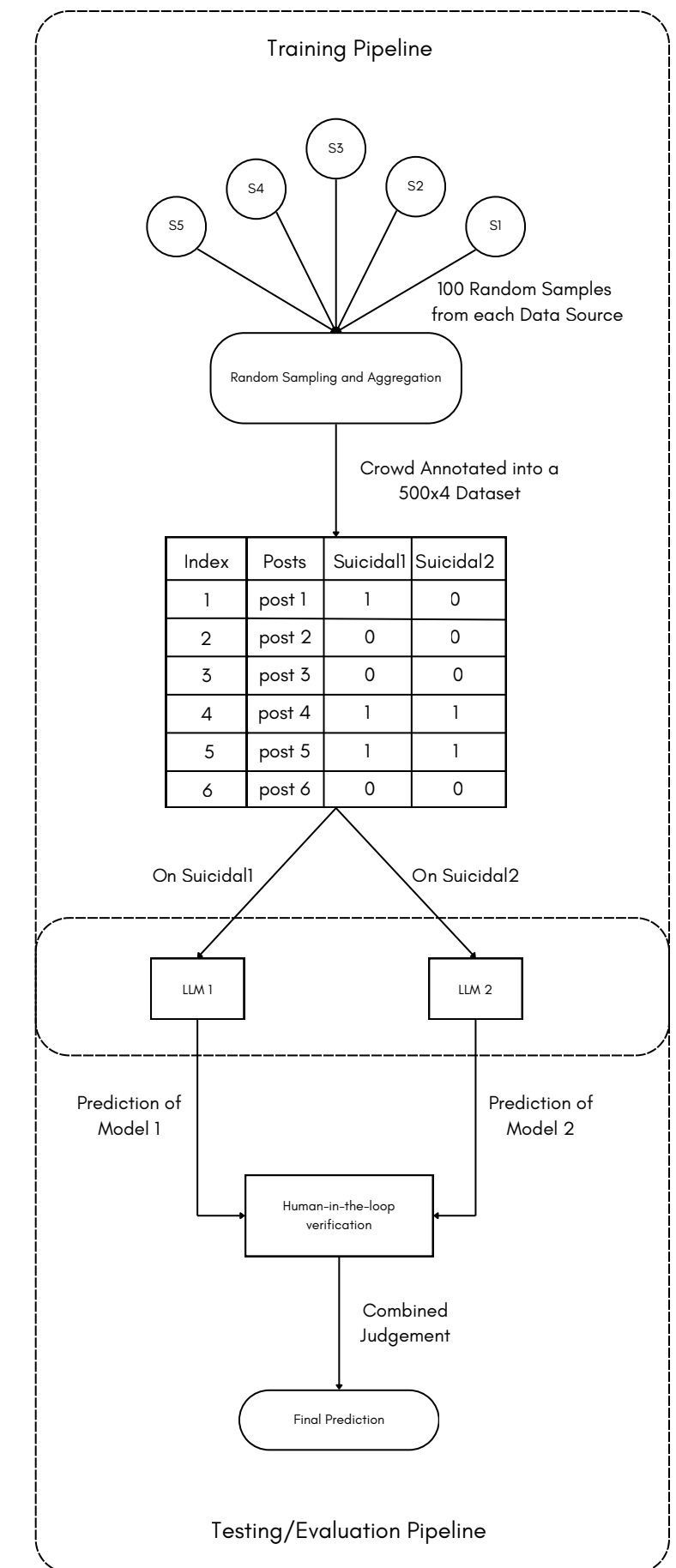


Fig 1. Dataset Creation, Training and Testing

DATASET CREATION OVERVIEW

Random Sampling and Aggregation (Phase 1):

The Dataset creation process in Phase 1 involved Random Sampling and Aggregation from 5 different open-source datasets containing user-generated textual posts from social media, to generate a customized 500 sample dataset. A small 100 sample set from this was manually annotated (Crowd Annotation) from two different sources independently resulting in two sets of annotations for each sample (Suicidal1 and Suicidal2).

Data Augmentation and Inter-Tagger Agreement Evaluation (Phase 3):

The remaining 400 samples of data were crowd annotated again and augmented to the previous set of 100 samples to generate a total set of 500 samples, for assessing inter-tagger agreement (using Kappa Score) and final model training.

Tweet	Suicide
making some lunch	Not Suicide post
@Alexia You want his money.	Not Suicide post
@dizzyhrvy that crap took me forever to	Potential Suicide post
@jnaylor #kiwitweets Hey Jer! Since wh	Not Suicide post
Trying out "Delicious Library 2&qu	Not Suicide post
@ValenValdez Oh, that's good to hear. E	Not Suicide post
@mcm180 u've got a list for fellow #hot	Not Suicide post

Fig 2. Suicide_Ideation_Dataset(Twitter-based)

+ Crowd Annotation →

Index	Posts	Suicidal1	Suicidal2
1	like what, for insta	0	0
2	Describe type pull	0	0
3	Tired of kickin you	0	0
4	yes, they are.	0	0
5	I thought which co	0	0

500 x 4

Fig 3. Suicidal Ideation Dataset

EXPLORATORY DATA ANALYSIS

Inter-Tagger Agreement using Cohen Kappa Score:

The Crowd Annotated 500 sample dataset was checked for the rate of agreement and disagreement among the Annotator Groups, ensuring a good rate of agreement while also capturing diverse sentiments of the same data.

- Kappa Score (κ) = 69.45% [Total Points of Disagreement = 44 out of 500]

Distribution of Suicidal & Non-Suicidal Posts (Intrinsic Biasness):

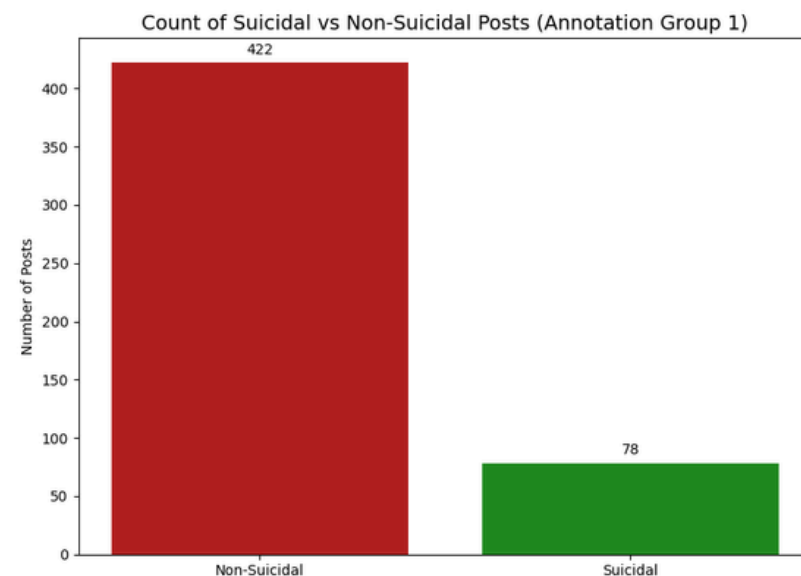


Fig 4. Non-Suicidal posts = 84.3%
Suicidal posts = 15.6%
(for Annotation-1)

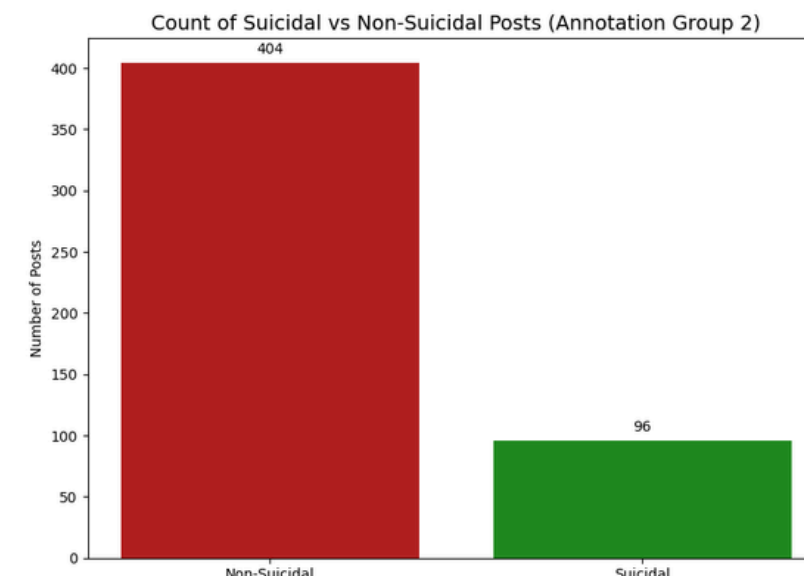


Fig 5. Non-Suicidal posts = 80.8%
Suicidal posts = 19.2%
(for Annotation-2)

Word Patterns in a Suicidal vs. Non-Suicidal Post (Word Clouds):



Fig 6. Annotation-1

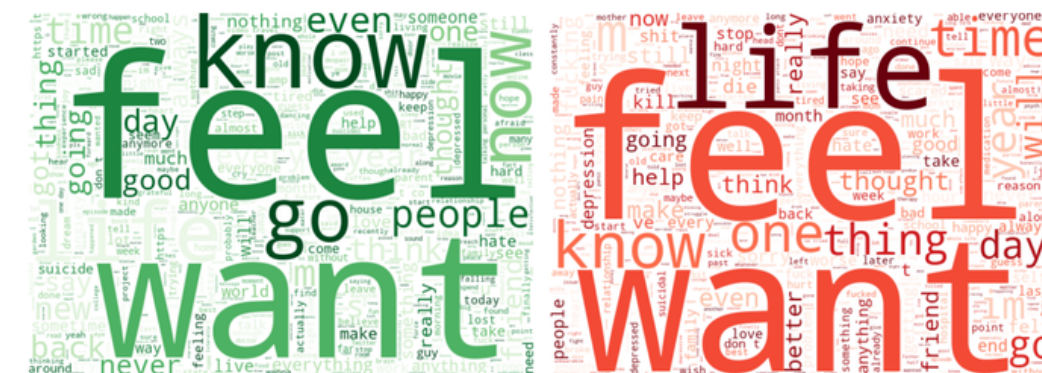


Fig 7. Annotation-2

LITERATURE REVIEW AND RESEARCH GAPS

Analysis of Previous Works:

Recent research has evolved from traditional NLP and Machine Learning (TF-IDF, SVM, RF) to hybrid Deep Learning (CNN-BiLSTM) and Transformer architectures (BERT, GPT-2, DistilBERT) [1], [2]. Significant advancements include the integration of lexicon-based sentiment scores and latent topics [3] to enhance text representation. While hybrid models like CNN-BiLSTM with triple embeddings [4] and zero-shot LLMs [5] achieve high benchmark performance (90–97% F1), they are often optimized on balanced Reddit and Twitter datasets [6], [7].

Research Gaps:

Current methodologies struggle significantly on various perspectives of the Suicidal Modelling, due to

- Models trained on artificially balanced datasets, failing to generalize on the intrinsic bias of the suicidal cases.
- Lack of labelled datasets validated from trusted human sources, capturing diverse real-world perspectives.
- Platform biasness as most research works focus on large datasets from Reddit or Twitter platforms only.
- AI snapshot predictions cannot account for long-term psychological ideation-to-action buildup.

How our Proposed Solution helps to mitigate these Gaps (Novelties):

- Models trained on practically biased dataset crowd annotated from diverse demographics (Human-in-the-loop), capturing varying real-world sentiments through valid labeling (validated by Kappa Score, κ).
- Datasets are randomly sampled and aggregated from diverse social media platforms.
- Hybrid decision making, based on temporal patterns recognized by humans and snapshot predictions made by AI.

MACHINE LEARNING - BASED MODELS

Overview:

In this project, three traditional Machine Learning classifiers - Logistic Regression (LR), Support Vector Machine (SVM), and Random Forest (RF) were used as baseline models for suicidal ideation detection. These models work effectively with high-dimensional TF-IDF features and offer fast training, good interpretability, and reliable performance for binary classification tasks such as Suicidal vs Non-Suicidal classification.

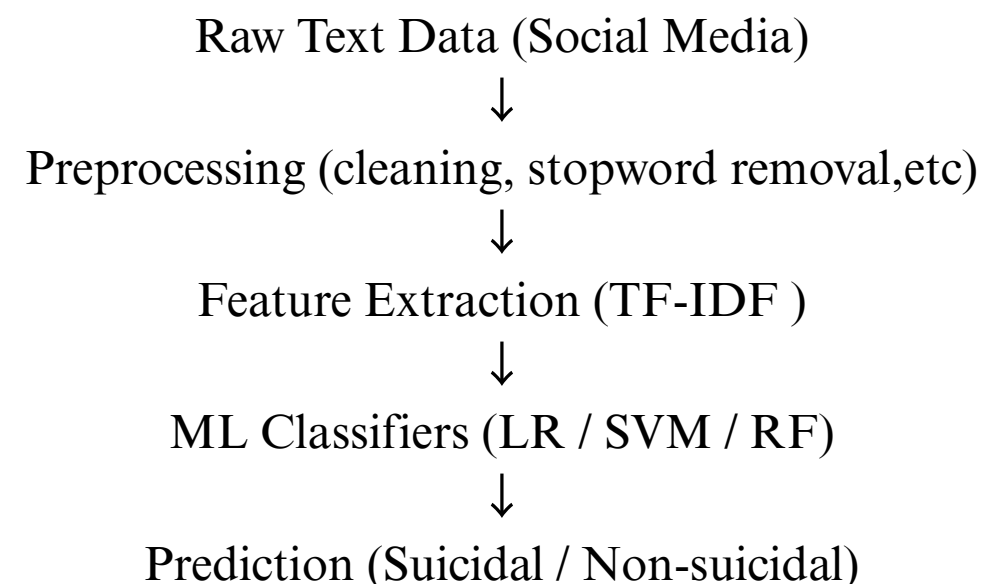
Implementation details:

The models were trained on TF-IDF vectorized text after applying data cleaning steps such as removing HTML tags, emojis, hashtags, and converting text to lowercase and removed extra punctuation mark. To address class imbalance, class weighting and stratified splitting were applied. Each classifier was tuned for better generalization:

- **Logistic Regression:** Fits a linear boundary and works well with large text features.
- **SVM:** Uses a maximum-margin hyperplane to separate the classes clearly.
- **Random Forest:** Uses many decision trees to capture complex, non-linear patterns.

These ML models provided strong baseline results.

Flow Diagram:



MACHINE LEARNING - BASED MODELS

Performance:

Model	Training Accuracy	Testing Accuracy
Logistic Regression	91%	84%
SVM	93%	86%
Random Forest	94%	86%

Table 1. Suicidal-1 Column Evaluation Metrics

Model	Training Accuracy	Testing Accuracy
Logistic Regression	90%	82%
SVM	91%	87%
Random Forest	92%	83%

Table 2. Suicidal-2 Column Evaluation Metrics

For Suicidal1, SVM and Random Forest gave the best testing accuracy.
For Suicidal2, SVM gave high training and testing accuracy.

MACHINE LEARNING - BASED MODELS

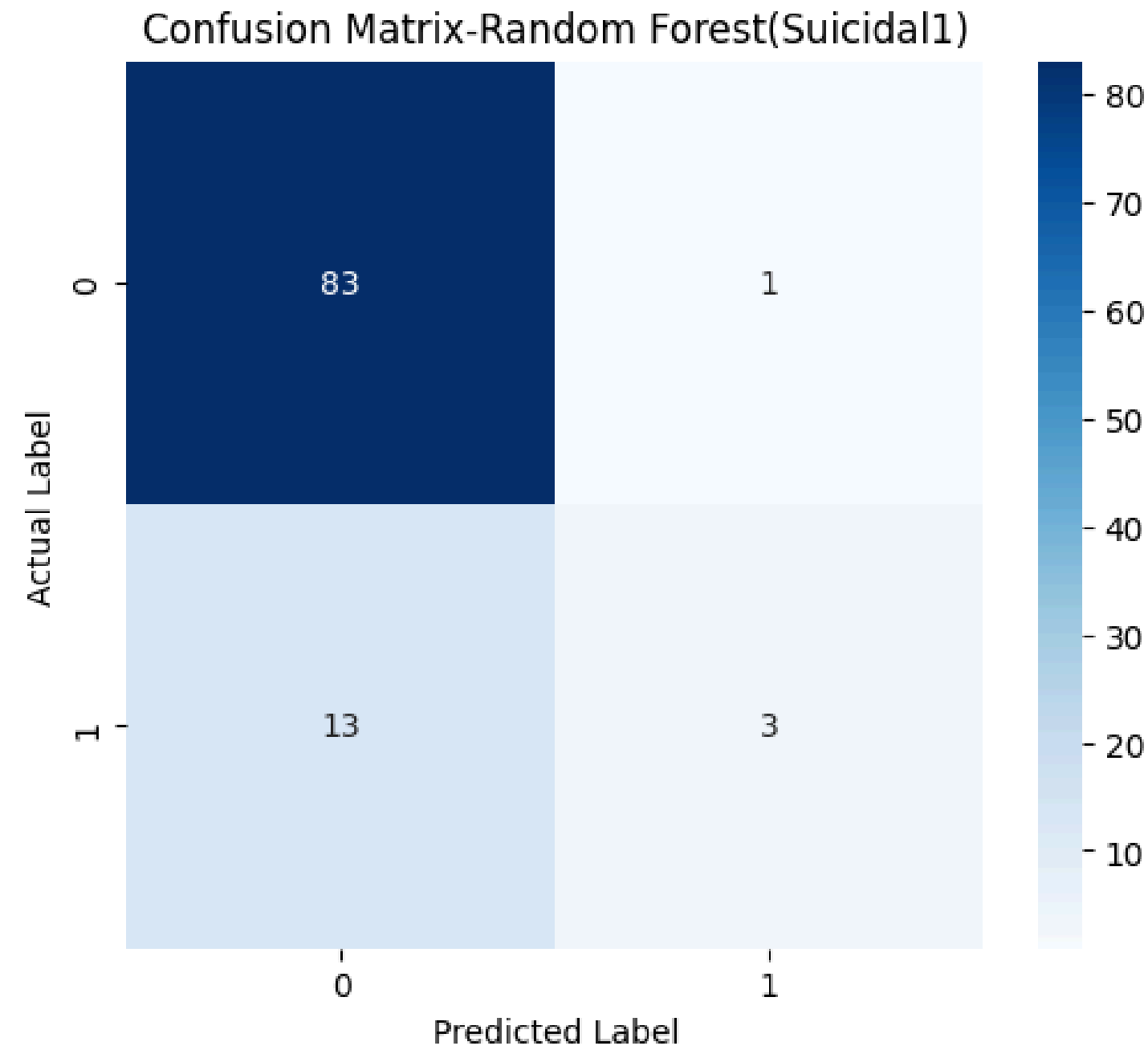


Fig 8. Confusion Matrix for Suicidal1

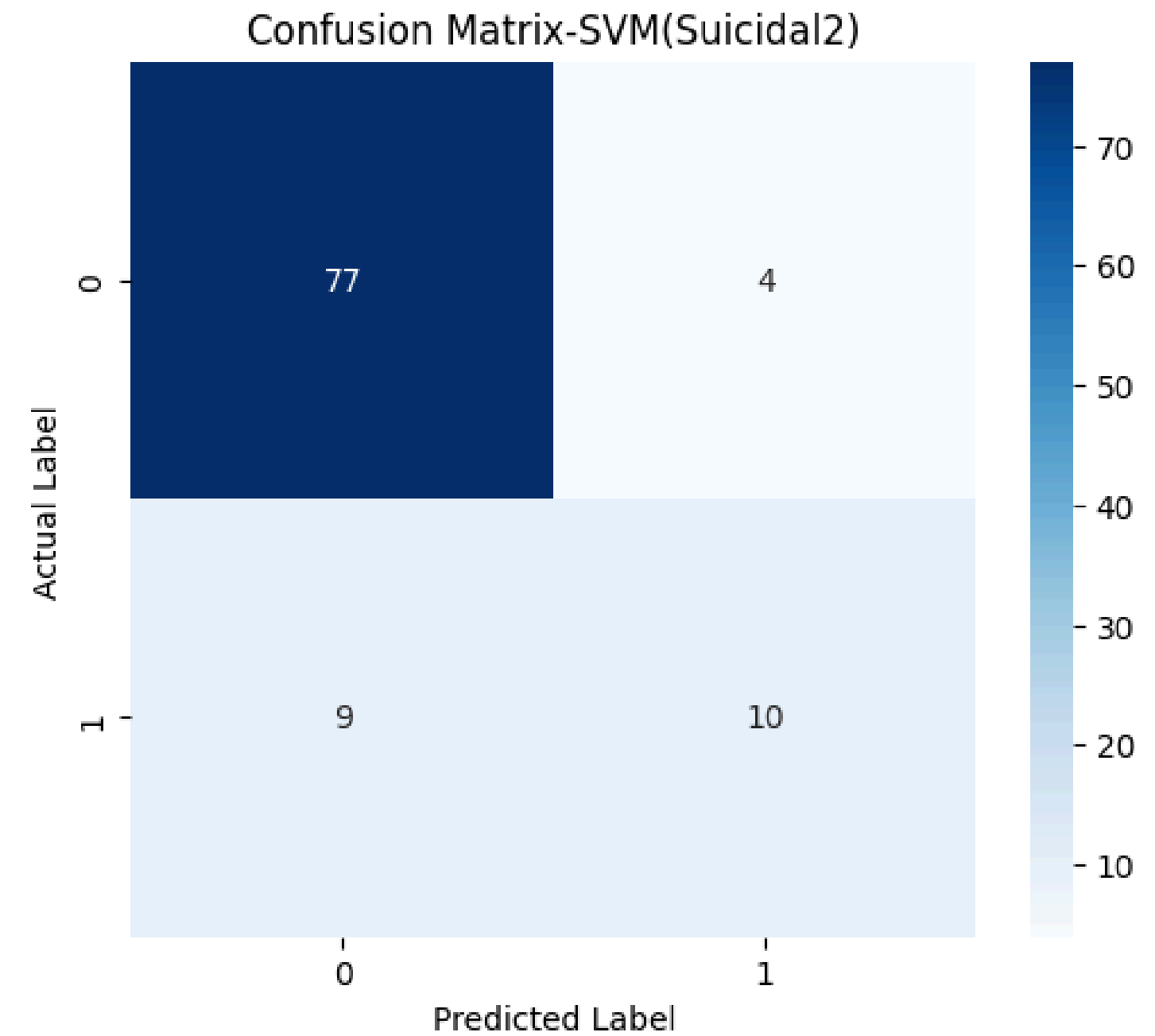


Fig 9. Confusion Matrix for Suicidal2

BERT MODEL

Overview:

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based language representation model developed by Google AI (Devlin et al., 2019). The "uncased" variant converts all input text to lowercase, making the model case-insensitive, which is beneficial for tasks where case information is not critical.

Implementation details:

In this project, BERT was fine-tuned for binary classification (Suicidal vs Non-Suicidal) using the Hugging Face Transformers library with GPU acceleration. The BERT tokenizer was used to preprocess the input, and all experiments were conducted using the standard implementation pipeline for efficient learning and robust performance.

Performance:

- Suicidal1 Accuracy: 89%
- Suicidal2 Accuracy: 91%

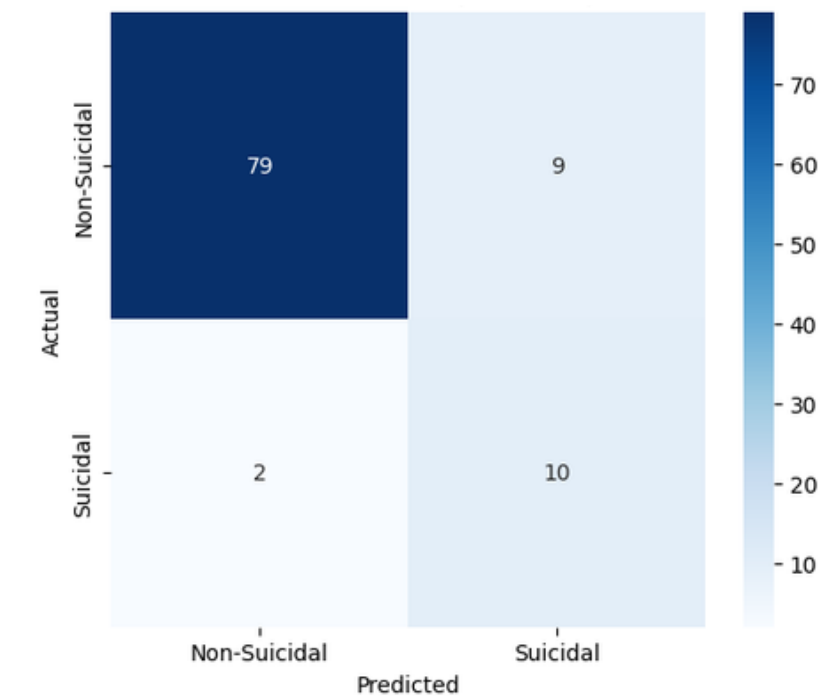


Fig 10. Confusion matrix for Suicidal1

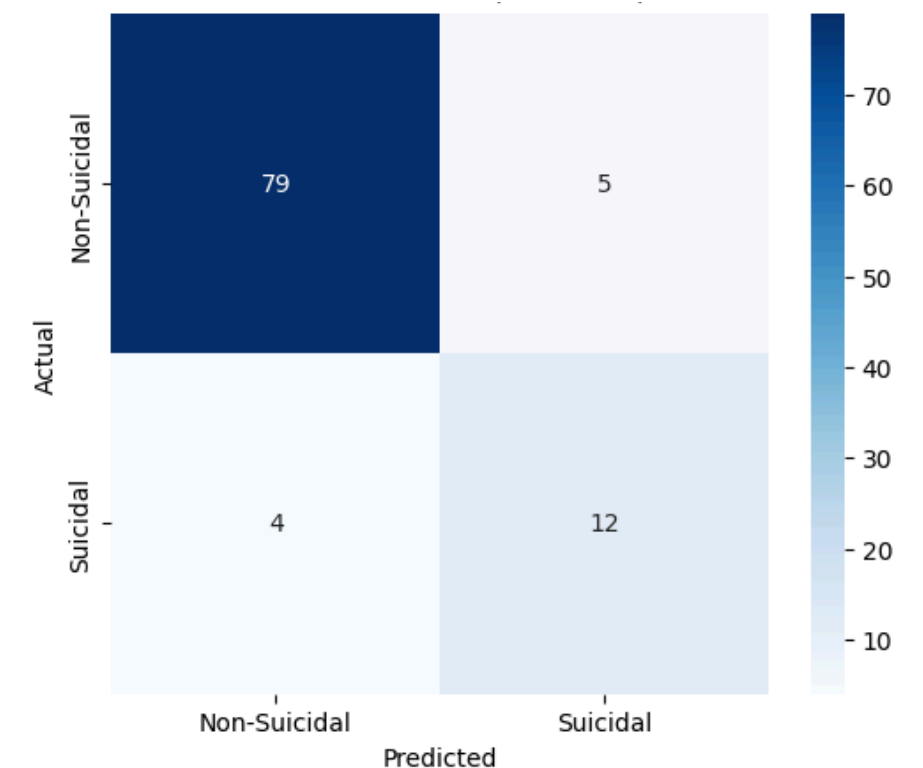


Fig 11. Confusion matrix for Suicidal2

ROBERTA MODEL

Overview:

RoBERTa (Robustly Optimized BERT Approach) is a transformer-based language model developed by Facebook AI. It is a transformer-based language model designed for understanding contextual relationships in text through bidirectional encoding.

Implementation details:

In this project, ROBERTa-base was fine-tuned for binary classification (Suicidal vs Non-Suicidal) using the Hugging Face Transformers library, leveraging GPU acceleration for efficient learning. It was selected for its strong results on sequence-level and token-level classification tasks, and effective detection of suicidal ideation in user posts.

Performance:

- Suicidal1 Accuracy: 96%
- Suicidal2 Accuracy: 91%

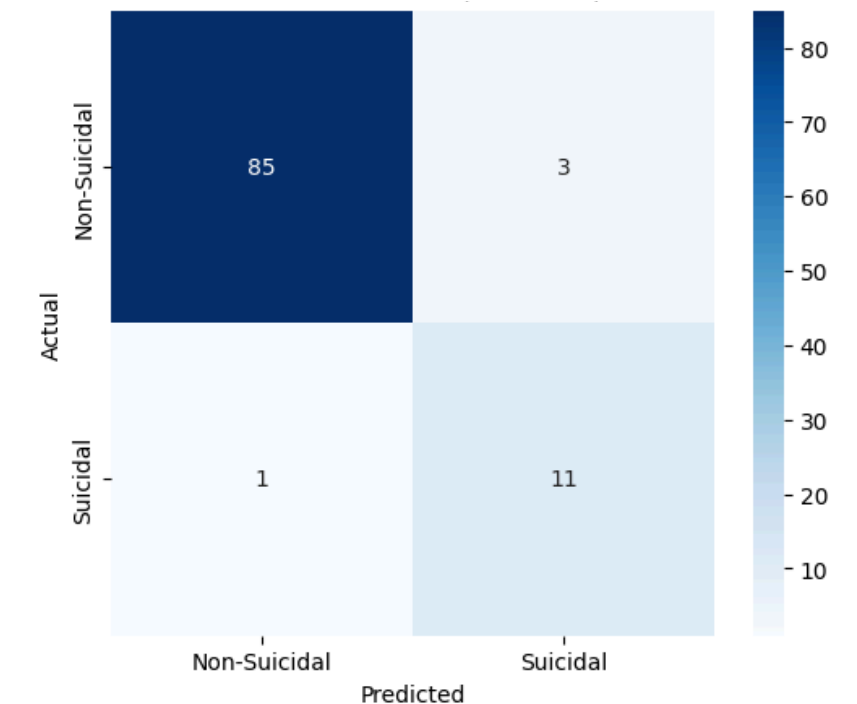


Fig 12. Confusion matrix for Suicidal1

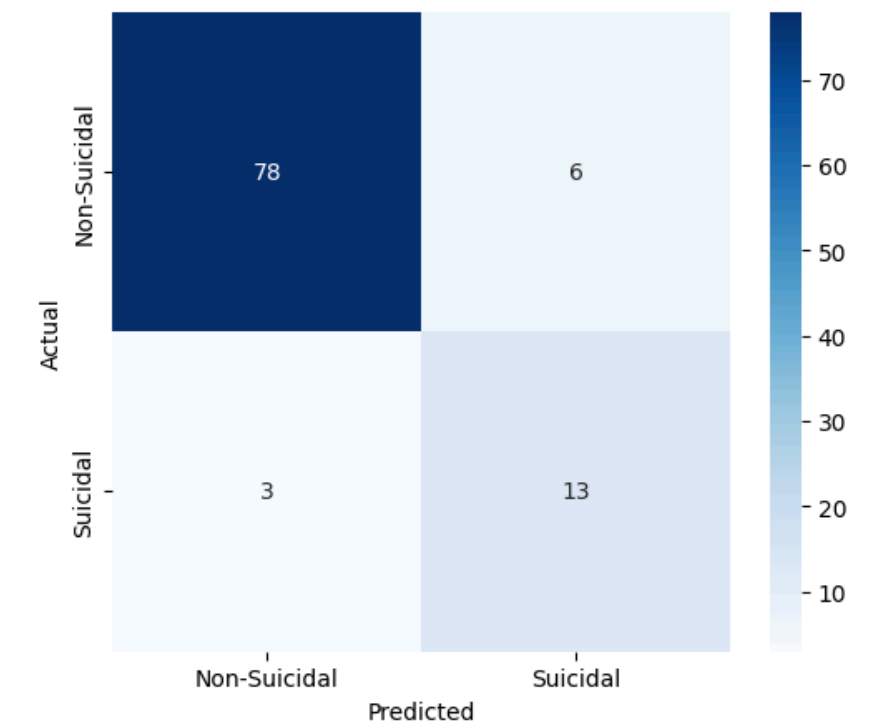


Fig 13. Confusion matrix for Suicidal2

DISTILBERT MODEL

Overview:

- Lightweight distilled version of BERT developed by Hugging Face.
- Designed to retain 97% of BERT's performance, 40% smaller and 60% faster.
- Uses a bi-directional Transformer encoder to capture deep contextual meaning in text.

Implementation details:

The model is fine-tuned for binary classification: Suicidal vs Non-Suicidal, using Hugging Face Transformers with GPU acceleration. Some of the specifications includes as follows:

- DistilBERT tokenizer for subword tokenization
- Max sequence length = 128
- AdamW optimizer with weight decay
- Cross-entropy loss for classification
- Early stopping based on validation loss

Performance:

- Suicidal1 Accuracy: 88%
- Suicidal2 Accuracy: 90%

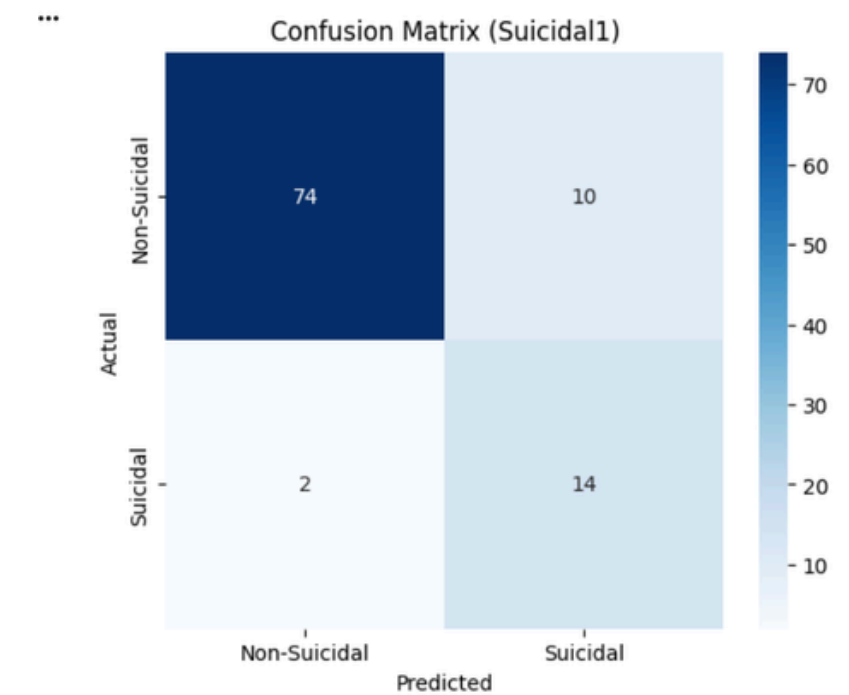


Fig 14. Confusion Matrix for Suicidal1

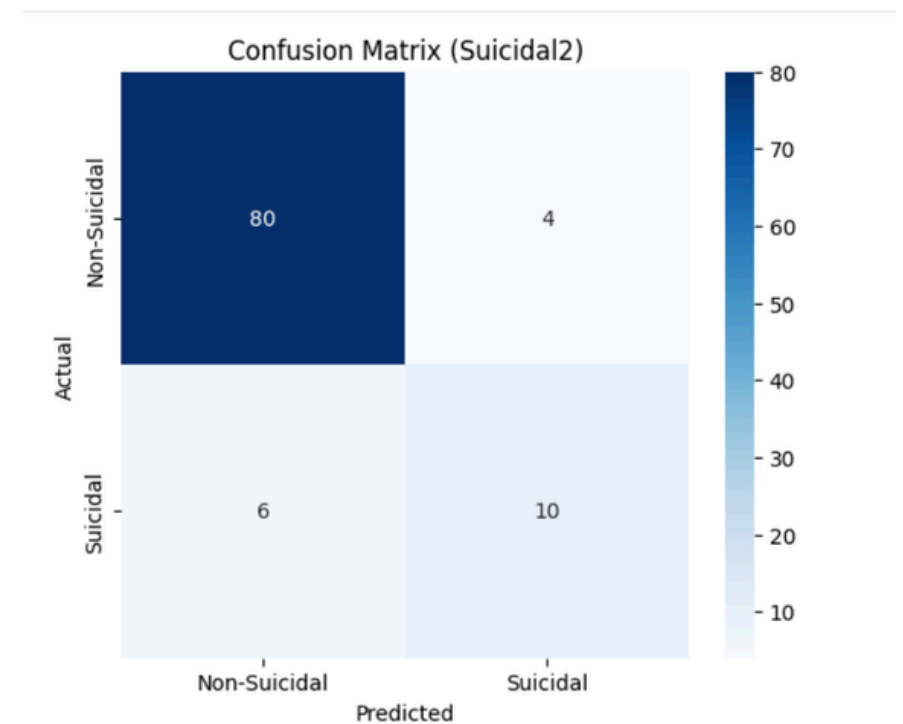


Fig 15. Confusion Matrix for Suicidal2

ALBERT MODEL

Overview:

ALBERT (A Lite BERT) is a parameter-efficient transformer-based language model developed by Google Research. It maintains the same bidirectional encoding capabilities as BERT but significantly reduces model size through techniques like factorized embeddings and cross-layer parameter sharing, enabling faster and more memory-efficient training without sacrificing performance.

Implementation details:

In this project, ALBERT-base-v2 was fine-tuned for binary classification (Suicidal vs Non-Suicidal) using the Hugging Face Transformers library with GPU acceleration. It was chosen due to its lightweight architecture, reduced parameter count, and strong performance on classification tasks, making it well-suited for detecting suicidal ideation in social media posts.

Performance:

- Suicidal1 Accuracy: 91%
- Suicidal2 Accuracy: 84%

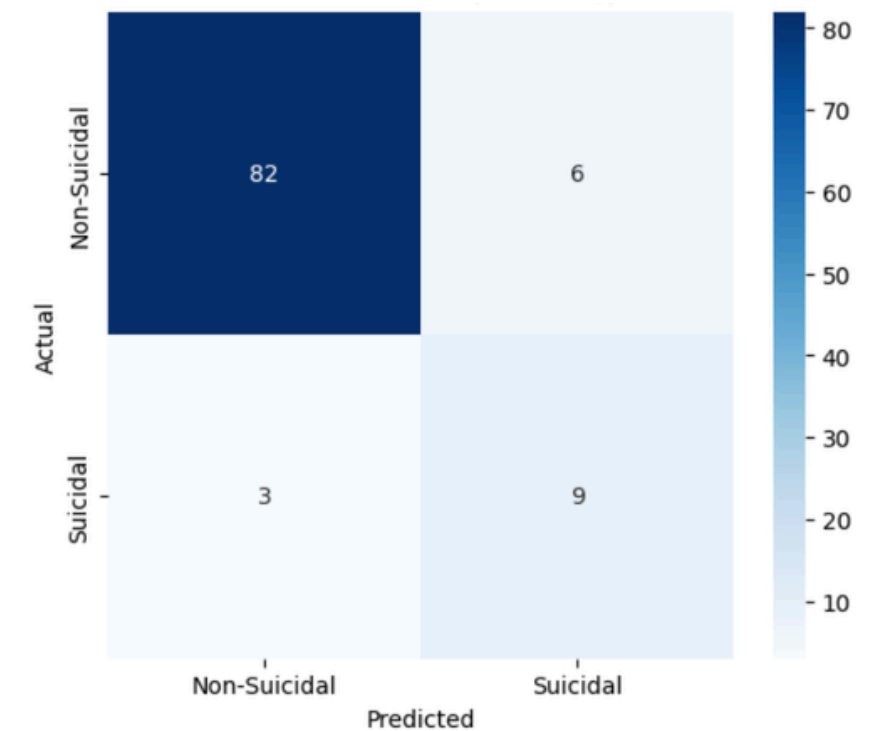


Fig 16. Confusion matrix for Suicidal1

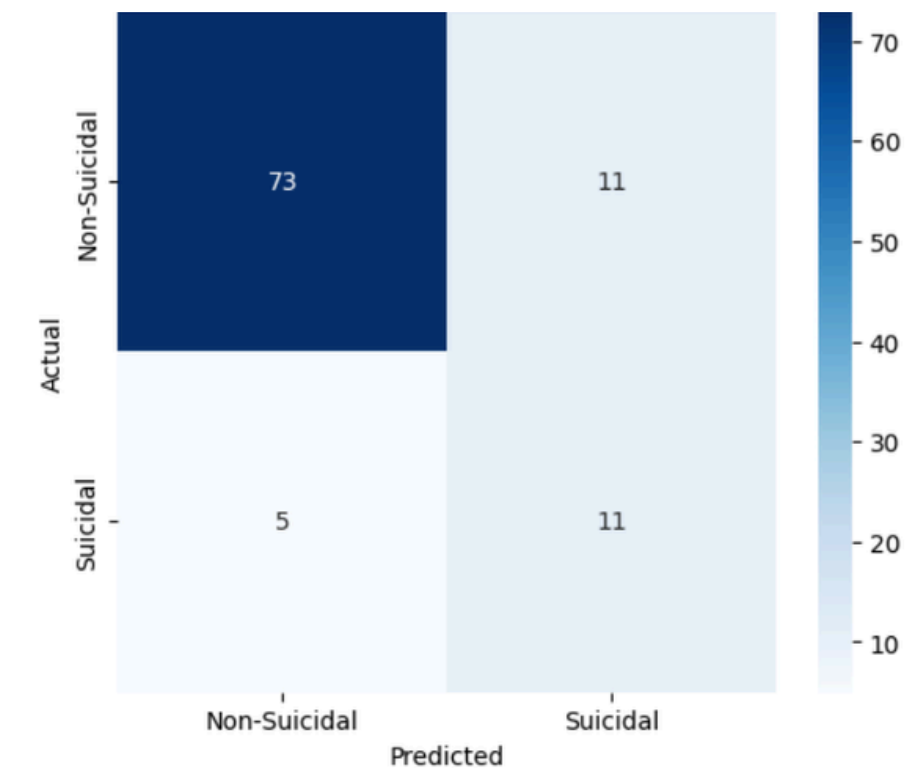


Fig 17. Confusion matrix for Suicidal2

ELECTRA MODEL

Overview:

ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) is a transformer-based language model developed by Google Research. Unlike BERT-style models that rely on Masked Language Modeling (MLM), ELECTRA uses a more efficient pre-training task called Replaced Token Detection (RTD), allowing the model to learn from every token rather than only the masked ones.

Implementation details:

In this project, ELECTRA was fine-tuned for binary classification (Suicidal vs Non-Suicidal) using Hugging Face Transformers and GPU acceleration. ELECTRA was chosen for its sample-efficient pre-training and strong contextual understanding, making it effective for identifying suicidal ideation in user posts.

Performance:

- Suicidal1 Accuracy: 89%
- Suicidal2 Accuracy: 86%

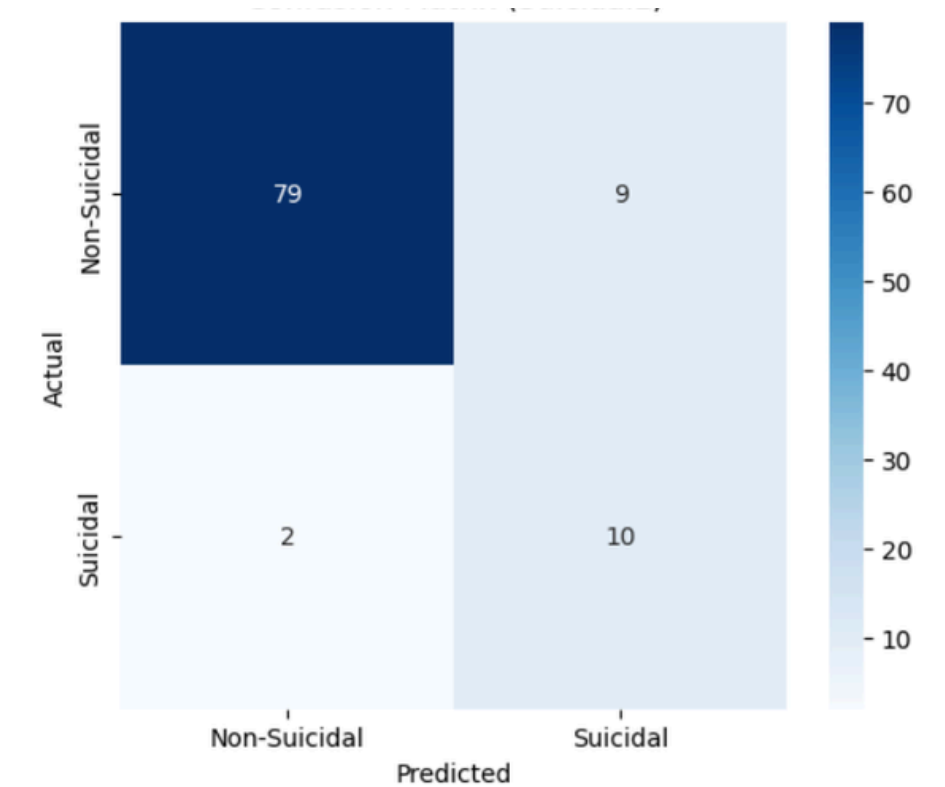


Fig 18. Confusion Matrix for Suicidal1

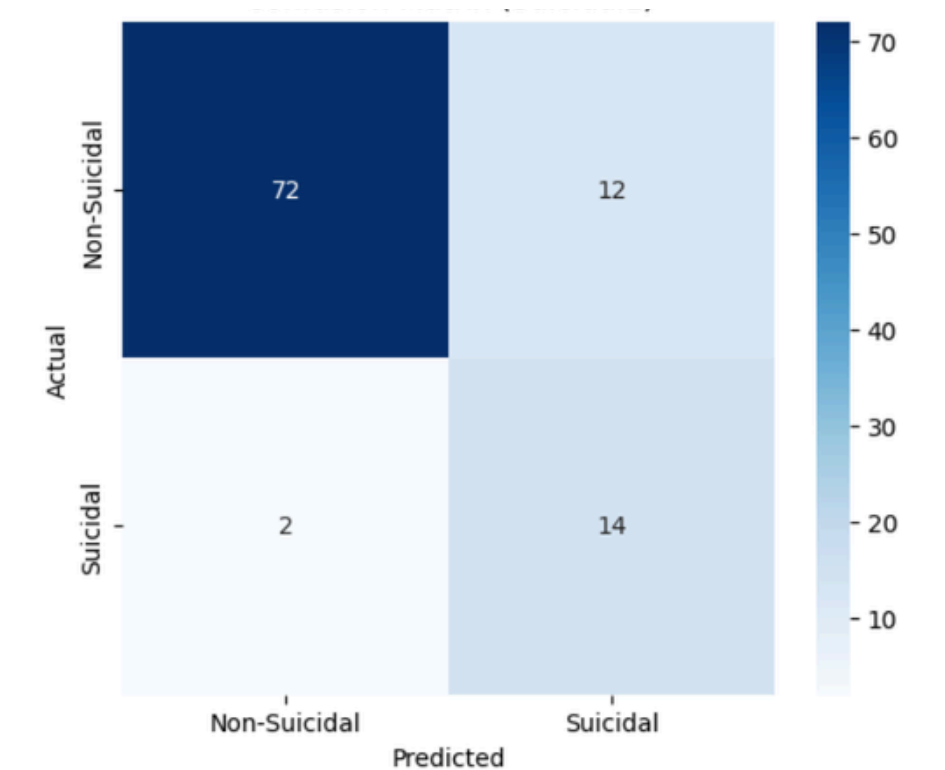


Fig 19. Confusion Matrix for Suicidal2

RESULT AND DISCUSSIONS

Model	Validation Accuracy	Validation Loss
DistilBERT	0.88	0.317
RoBERTa	0.96	0.2518
BERT-base-uncased	0.89	0.3189
ALBERT-base-v2	0.91	0.3361
ELECTRA-base	0.89	0.3304
Logistic Regression	0.84	0.3824
SVM	0.86	0.2962
Random Forest	0.86	0.3503

Table 3. Model Performances for Annotation-1 Column

Model	Validation Accuracy	Validation Loss
DistilBERT	0.90	0.284
RoBERTa	0.91	0.3165
BERT-base-uncased	0.91	0.3656
ALBERT-base-v2	0.84	0.5313
ELECTRA-base	0.86	0.3885
Logistic Regression	0.82	0.4216
SVM	0.87	0.3142
Random Forest	0.83	0.4813

Table 4. Model Performances for Annotation-2 Column

Results and Performance Analysis (Annotation-1 & Annotation-2):

RoBERTa achieves the highest validation accuracies of 96% and 91% for both Annotation-1 and Annotation-2 respectively, with lowest validation losses (0.2518 & 0.31565 respectively), which suggests the model generalizes well and minimizes errors.

PROOF OF CONCEPT AND CONCLUSION

The need for the proposed Hybrid Intelligence Framework:

The following examples demonstrates a small proof-of-concept of our intended Hybrid Intelligence framework, and how ensembled AI judgement and Human Judgement will work in coordination, complementing each other for accurate results.

Posts from Social Media	Estimator 1 (RoBERTa)	Estimator 2 (RoBERTa)	Human Estimation	Ground Truth
I am feeling very overwhelmed and hopeless lately.	0	0	0	0
I have nothing left to live fr... i just want to end this and sleep forever!	1	1	1	1
I'm not planning anything, but if a car suddenly swerved toward me today, I wouldn't move out of the way.	0	0	1	1
I am moving to the edge of the cliff, i have to do this for the betterment of others... i have to jmp frm here, i shall join the squad back after i respawn.	0	0	1	1
I want to die.	1	1	1	1
Just another bad day at work!	0	0	0	0
meet me at the battle arena, if i can't prove as MVP i shall commit suicide...	1	0	0	0
Broo! frr what u say this???? i hrd SSR commited suicide!	0	0	0	0
everyday i start out, then i cry my heart out, lonely is a man without love... - suits so damn well, but this phrasal life is becoming too suffocating now and i don't think i can make myself work with this on this world anymore... its just too much! when u feel such words depressing, uk the end is near :)	1	1	1	1
Hybrid Intelligence Framework Performance	6/9	7/9	9/9	—

Table 5. Proposed Hybrid Intelligence Framework with multiple estimators (Proof of Concept)

FUTURE SCOPE OF THE WORK

While the concluding evaluation on validation set has identified the best-performing models, the ultimate best model will be determined through real-life testing unseen datasets.

The final Model selection process would include:

- Cross-validation on properly annotated unseen datasets from verified clinical and psychological sources.
- Performance evaluation under diverse demographic and linguistic conditions.
- Ethical validation ensuring fairness, bias mitigation, and clinical reliability.

Proposed Future Implementations:

- To develop an actual Social Media Interface to test the generalization of this Project in a real life setup facing actual social media based interactions.
- Increase the volume of Training Data capturing more diverse patterns about social media sentiments through clinically validated annotations from Psychologists and applying the proposed Hybrid Intelligence framework in real life to test effectiveness.
- Design a robust and efficient Mitigation Strategy for alarming situations, like GPS Tracking, Immediate dial up at 100 with location share, Psychologist consulting, etc.

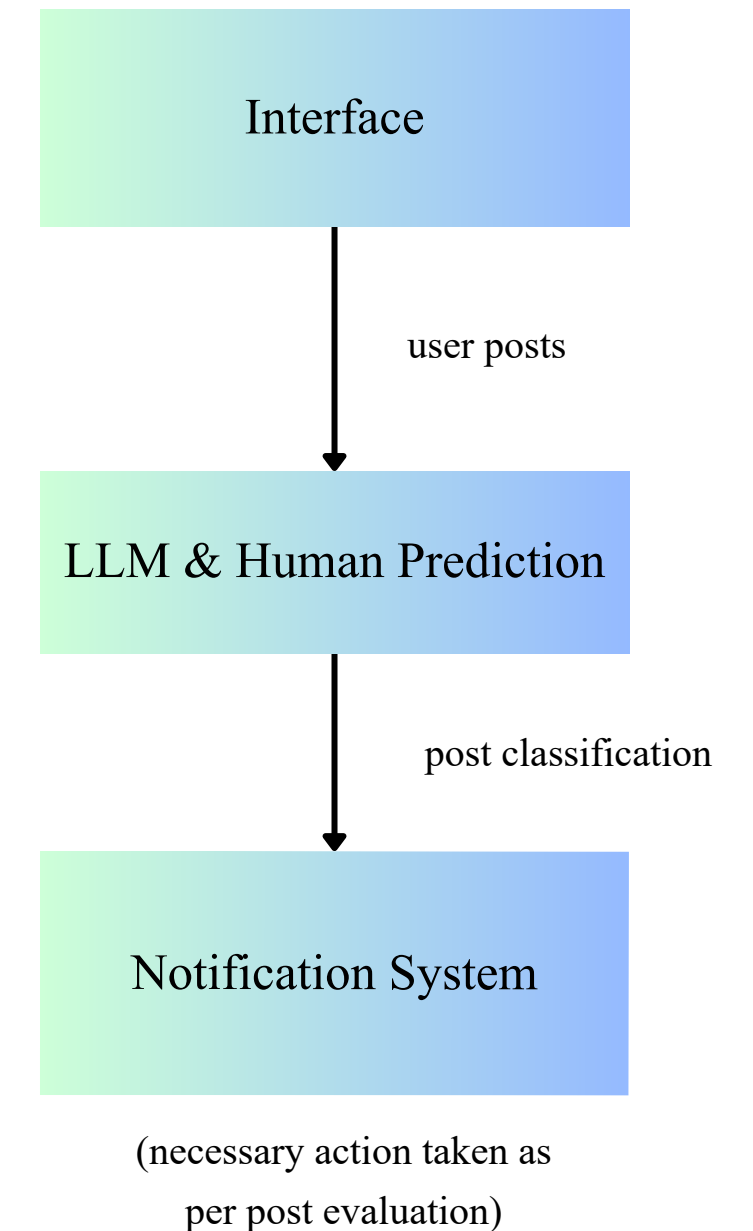


Fig 20. Final Workflow Diagram after the proposed Integration

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THANK YOU